**sPHENIX Internal Note**

# Vertex Distribution and $\Delta z$ in Single vs. Double-Interaction MC

Isolated Photon Cross-Section in $p$+$p$ at $\sqrt{s} = 200$ GeV

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**Abstract**

This note characterizes the vertex distributions and vertex displacement ($\Delta z = z_{\rm vtx}^{\rm reco} - z_{\rm vtx}^{\rm truth}$) in single-interaction and full GEANT double-interaction Monte Carlo samples used for the sPHENIX isolated photon cross-section measurement. In double-interaction events, the MBD-reconstructed vertex is the average of two collision vertices, producing a $\Delta z$ RMS of 48.1 cm (vs. 16.9 cm for single interaction, before vertex cut). The vertex displacement maps to a pseudorapidity shift via the EMCal barrel geometry: $\Delta\eta \approx -\Delta z/R_{\rm CEMC}$, broadening the cluster $\Delta\eta$ distribution by a factor of 5 (RMS 0.365 vs. 0.073). We document the vertex distribution shapes, the $\Delta z$ distribution properties, and the geometric origin of the $\Delta\eta$ correlation for both interaction types. These results inform the truth-matching strategy and pileup systematic uncertainty for the cross-section measurement.

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# 1 Introduction

In $p$+$p$ collisions at RHIC, the probability of two independent hard-scattering events occurring in the same bunch crossing (double interaction, or pileup) depends on the instantaneous luminosity and crossing angle. For Run 24 at $\sqrt{s} = 200$ GeV, the double-interaction cluster-weighted fractions are 22.4% at 0 mrad crossing angle and 7.9% at 1.5 mrad (event-level: 11.1% and 3.9%; computed by `calc_pileup_range.C`).

When two collisions occur in the same bunch crossing, the MBD (Minimum Bias Detector) reconstructs a single vertex that represents an average of the two true collision vertices. This displacement between the reconstructed and true (primary photon) vertex propagates to the cluster pseudorapidity via the EMCal barrel geometry:

$$\Delta\eta = \eta_{\text{cluster}} - \eta_{\text{truth}} \approx -\frac{\Delta z}{R_{\text{CEMC}} \cdot \cosh\eta} \tag{1}$$

where $R_{\text{CEMC}} = 93.5$ cm is the EMCal barrel radius.

Understanding the vertex distribution and its consequences is essential for three aspects of the measurement:

1. **Truth matching:** Vertex displacement drives $\Delta R$ matching failures (documented separately in Ref. [1]).
2. **Kinematic migration:** The $\eta$-shift changes both the reconstructed $E_{\text{T}} = E/\cosh\eta$ and the cluster's position relative to fiducial boundaries, affecting efficiency and the response matrix.
3. **Shower shape distortion:** Shifted kinematics alter the BDT score evaluation at the wrong working point of the $E_{\text{T}}$-dependent threshold.

## References


[1] sPHENIX Internal Note: Study of the $\Delta R$ Truth-Matching Requirement in Single-Interaction MC, 2025.


# 2 Samples and Method

The study uses the macro `study_vertex_double_interaction.C`, processing two `photon10` MC samples (truth photon $14 \leq p_{\text{T}} < 30$ GeV):

- **Single-interaction:** `photon10/bdt_split.root` — standard production MC with one collision per event.
- **Double-interaction:** `photon10_double/bdt_split.root` — full GEANT4 MC with two independent collisions simulated per event.

Two million events are processed per sample. Two classes of histograms are produced:

1. **Event-level** (filled before vertex cut): vertex distributions ($z_{\text{vtx}}^{\text{reco}}$, $z_{\text{vtx}}^{\text{truth}}$), $\Delta z$ distributions, and reco-vs-truth 2D correlation.
2. **Per-cluster** (filled after vertex cut and truth selection): $\Delta\eta$, $\Delta R$, $E_{\text{T}}$ ratio, and 2D $\Delta\eta$ vs. $\Delta z$ correlation for trkID-matched truth photons satisfying the standard PPG12 selection (`pid` $= 22$, `photonclass` $< 3$, $E_{\text{T}}^{\text{iso,truth}} < 4$ GeV, $|\eta| < 0.7$, $8 \leq p_{\text{T}} < 36$ GeV, $|z_{\text{vtx}}^{\text{reco}}| < 60$ cm).

The distinction matters: event-level distributions include all events (even those without selected truth photons or outside the vertex cut), while per-cluster distributions reflect the analysis-relevant population.

# 3 Results

## 3.1 Reconstructed Vertex Distributions

Figure 1 compares the reconstructed vertex distributions. The double-interaction reco vertex is *narrower* than single-interaction, which is a consequence of vertex averaging:

$$z_{\mathrm{vtx}}^{\mathrm{reco}} \approx \frac{z_1 + z_2}{2} \tag{2}$$

Averaging two independent draws from the beam vertex distribution reduces the variance by $\sqrt{2}$, producing a more peaked reco distribution.

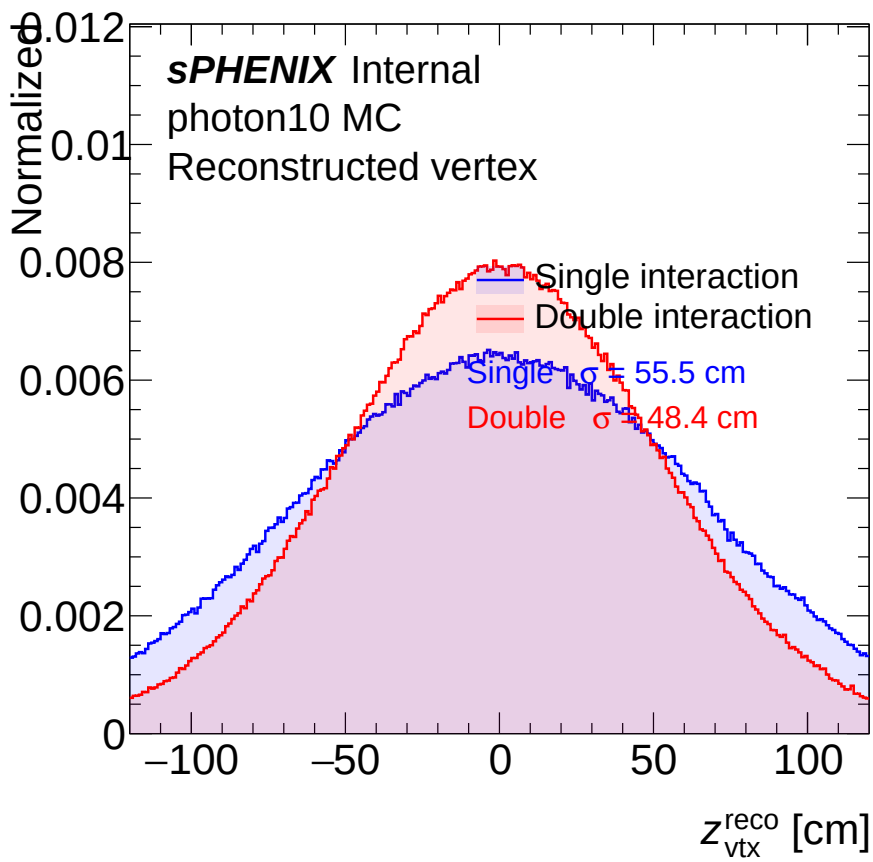


Figure 1: Normalized reconstructed vertex distributions for single-interaction (blue) and double-interaction (red) `photon10` MC. The double-interaction distribution is narrower due to the averaging of two collision vertices. Both are shown before the $|z_{\mathrm{vtx}}| < 60$ cm analysis cut.

### 3.2 Truth Vertex Distributions

The truth vertex (primary collision vertex) distributions are identical in shape for both samples, as expected — the beam profile is a property of the accelerator, not the number of interactions. Both follow the RHIC luminous region profile at $\sqrt{s} = 200$ GeV.

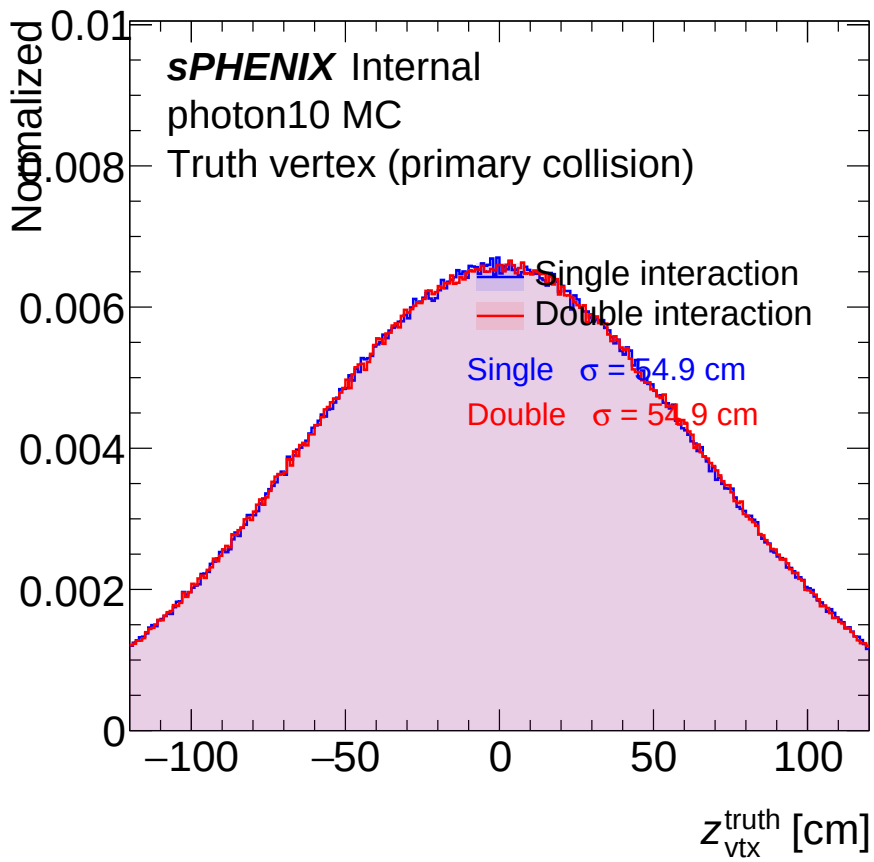


Figure 2: Normalized truth vertex distributions (primary collision vertex) for single and double-interaction MC. The distributions are consistent, reflecting the same RHIC beam profile.

### 3.3 Reco vs. Truth Vertex Correlation

Figure 3 shows the 2D correlation between reconstructed and truth vertex positions. In single-interaction MC, the correlation is tight along the diagonal with occasional off-diagonal artifacts from MBD vertex reconstruction failures. In double-interaction MC, the correlation band is significantly broadened: the reco vertex is pulled toward the second collision vertex, away from the truth.

### 3.4 Vertex Displacement ($\Delta z$) Distributions

Figure 4 shows the $\Delta z$ distribution for both samples. The single-interaction distribution has a sharp central spike (well-reconstructed events) with non-Gaussian tails extending to $|\Delta z| > 100$ cm from MBD reconstruction failures. The double-interaction distribution is broad and approximately Gaussian with heavy tails, reflecting the convolution of two independent beam vertex distributions.

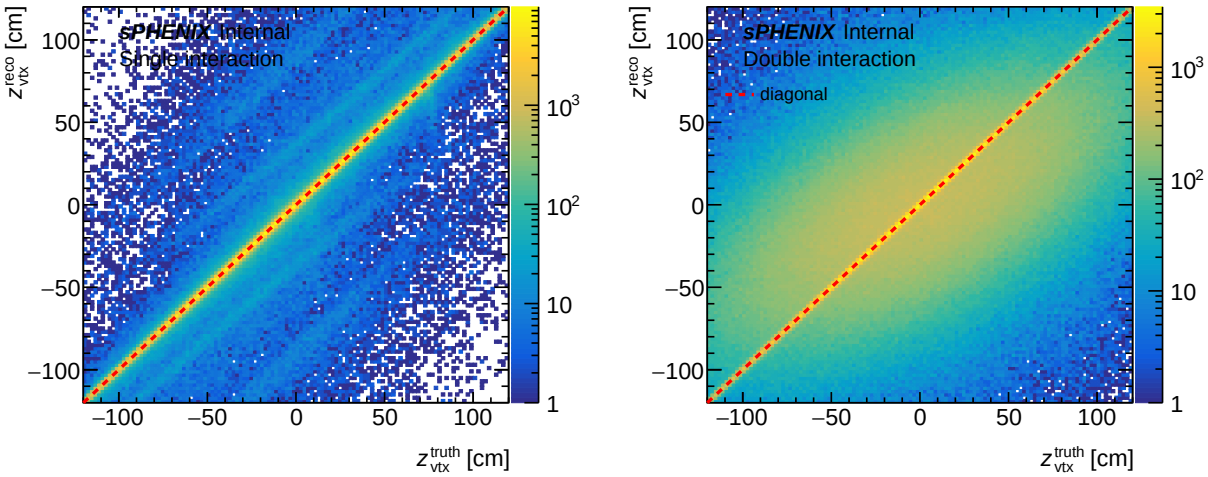


Figure 3: Two-dimensional correlation between reconstructed and truth vertex positions for single-interaction (left) and double-interaction (right) MC. The red dashed line shows the diagonal ($z_{\mathrm{vtx}}^{\mathrm{reco}} = z_{\mathrm{vtx}}^{\mathrm{truth}}$). The double-interaction sample shows significant broadening around the diagonal due to the second collision vertex.

Table 1: Vertex displacement statistics (event-level, before vertex cut).

| **Metric** | **Single** | **Double** |
|---|---|---|
| $\Delta z$ mean | $-0.16$ cm | $+0.13$ cm |
| $\Delta z$ RMS | 16.9 cm | 48.1 cm |
| Fraction $\lvert\Delta z\rvert > 9.35$ cm | 36.0% | 74.2% |

The critical threshold $|\Delta z|_{\text{crit}} = \Delta R_{\text{cut}} \times R_{\text{CEMC}} = 0.1 \times 93.5 = 9.35$ cm marks where the vertex-induced $\Delta\eta$ exceeds the $\Delta R = 0.1$ matching cut (at $\eta = 0$). In double-interaction MC, 74.2% of events exceed this threshold, compared to 36.0% in single-interaction (before vertex cut). The large single-interaction fraction is dominated by events outside the $|z_{\text{vtx}}| < 60$ cm cut; within the analysis acceptance, the effective $\Delta R$ failure rate drops to 3.04%.

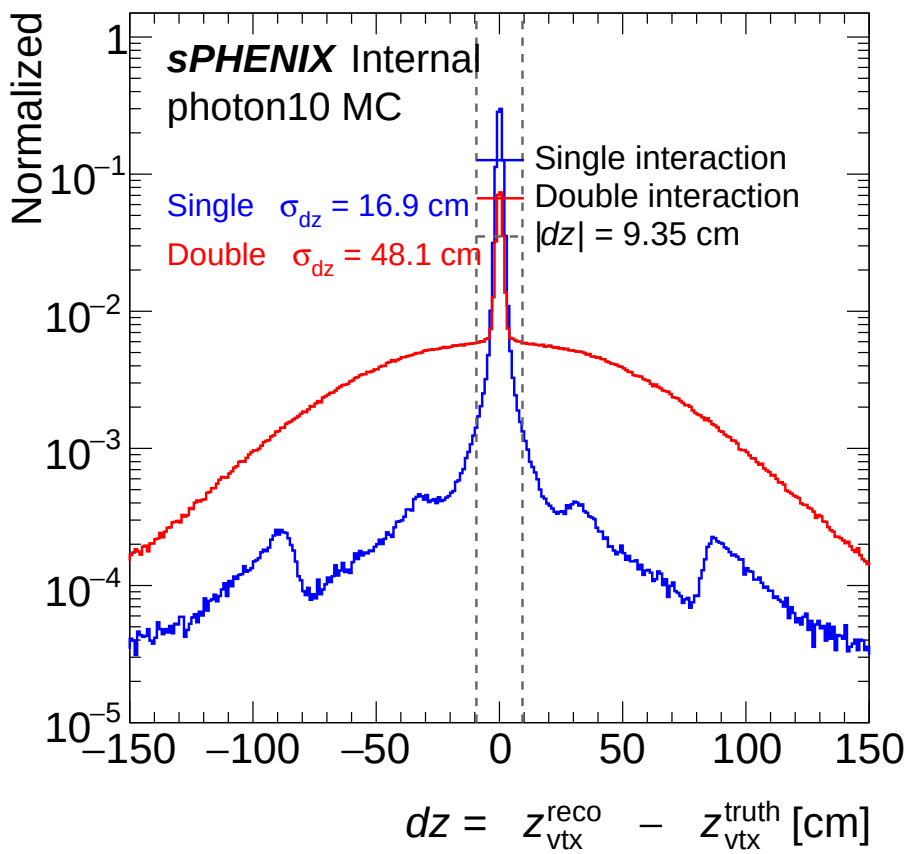


Figure 4: Normalized $\Delta z$ distributions for single (blue) and double (red) interaction MC. Dashed lines mark the critical threshold $|\Delta z| = 9.35$ cm. The double-interaction distribution is dramatically broader, with most events exceeding the threshold.

### 3.5 Physics of the $\Delta z$ Distribution

For double-interaction events, if both collision vertices are drawn independently from the beam luminous region with distribution $f(z)$ (approximately Gaussian, $\sigma_z \approx 30$ cm for RHIC at $\sqrt{s} = 200$ GeV), then:

$$\Delta z = z_{\text{vtx}}^{\text{reco}} - z_1 \approx \frac{z_1 + z_2}{2} - z_1 = \frac{z_2 - z_1}{2} \tag{3}$$

The distribution of $(z_2 - z_1)$ for two independent draws from $f(z)$ has variance $2\sigma_z^2$, so:

$$\sigma_{\Delta z}^{\text{double}} = \frac{\sqrt{2}\,\sigma_z}{2} = \frac{\sigma_z}{\sqrt{2}} \tag{4}$$

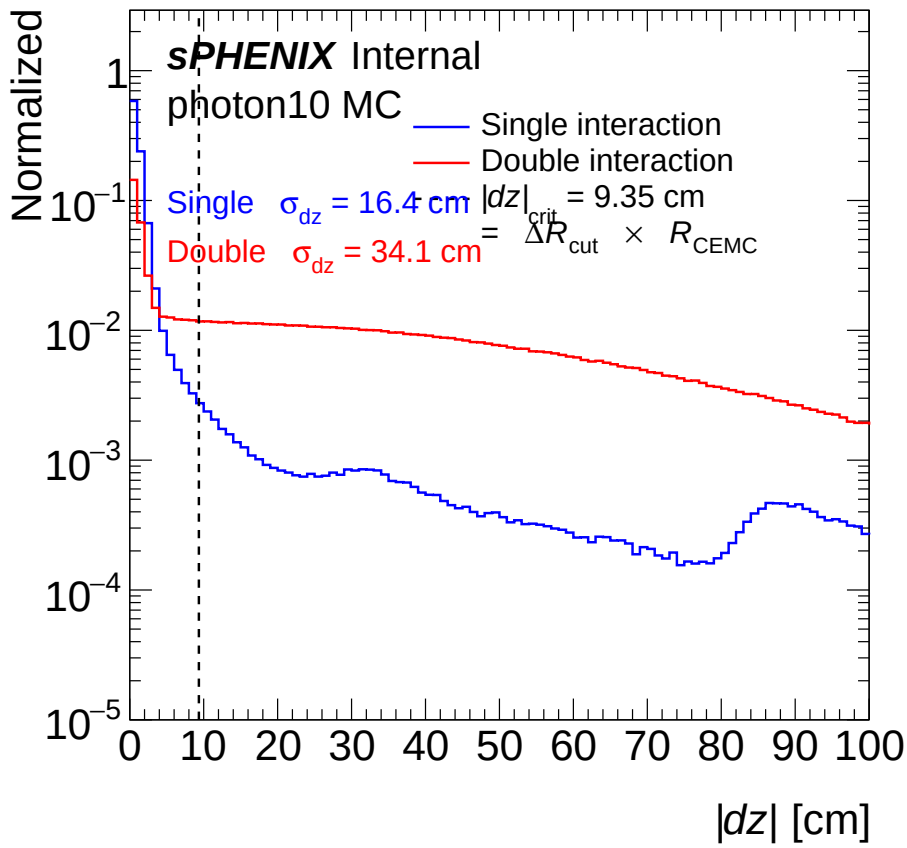


Figure 5: Normalized $|\Delta z|$ distributions. The single-interaction distribution drops steeply beyond 5 cm, while the double-interaction distribution remains approximately flat out to ∼60 cm. The critical threshold $|\Delta z|_{\mathrm{crit}} = 9.35$ cm is marked.

With $\sigma_z \approx 30$ cm, the predicted $\sigma_{\Delta z} \approx 21$ cm. The measured value of 48.1 cm is larger, reflecting the non-Gaussian tails of the RHIC luminous region and the contribution of events where the MBD vertex reconstruction is biased beyond simple averaging.

For single-interaction events, $\Delta z$ arises solely from MBD vertex reconstruction resolution. The sharp central peak ($\sigma \sim 3$ cm) represents well-reconstructed events, while the tails reflect events with insufficient MBD timing information for precise vertex determination.

### 3.6 $\Delta\eta$ vs. $\Delta z$ Correlation

Figure 6 shows the 2D correlation between cluster $\Delta\eta$ and vertex displacement $\Delta z$ for double-interaction MC. The tight linear correlation confirms the geometric origin (Eq. 1), with the finite band width arising from the $\cosh\eta$ factor varying across the barrel ($|\eta| < 0.7$).

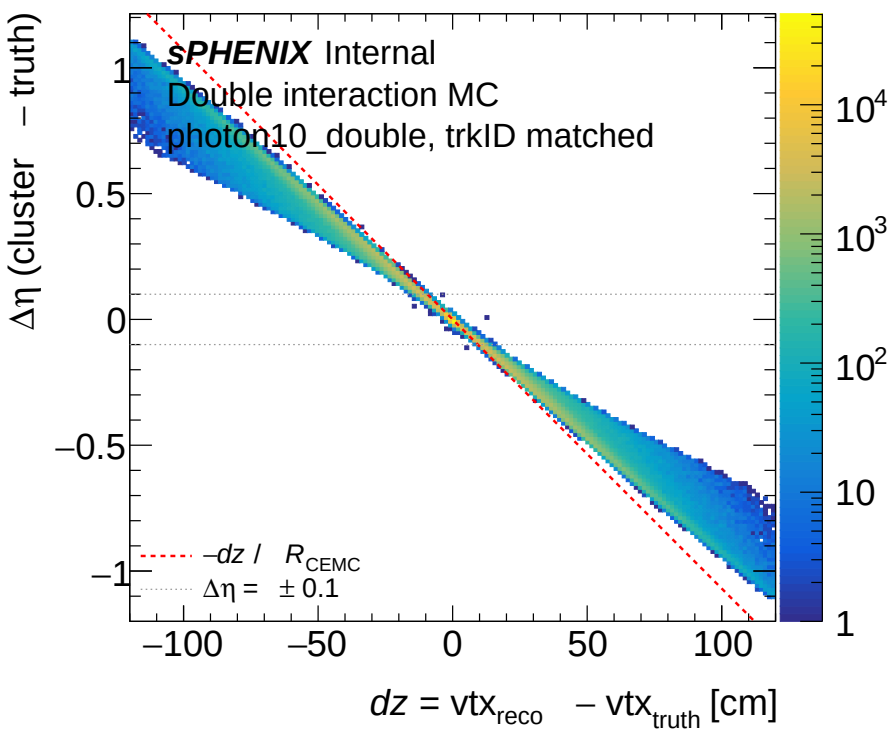


Figure 6: Two-dimensional $\Delta\eta$ vs. $\Delta z$ for double-interaction MC (`photon10_double`, trkID-matched truth photons). The red dashed line shows $\Delta\eta = -\Delta z/R_{\mathrm{CEMC}}$ at $\eta = 0$. Dotted lines mark $\Delta\eta = \pm 0.1$, corresponding to the $\Delta R$ matching threshold at $\Delta\phi \approx 0$. The band width reflects the $\cosh\eta$ variation across the barrel acceptance.

### 3.7 $\Delta\eta$ Distribution

Figure 7 compares the $\Delta\eta$ distributions. The single-interaction distribution is narrowly peaked (RMS 0.073), while the double-interaction distribution extends to $|\Delta\eta| > 1.0$ (RMS 0.365, a factor of 5 broader). This broadening directly propagates to $E_{\mathrm{T}}$ migration and BDT threshold evaluation.

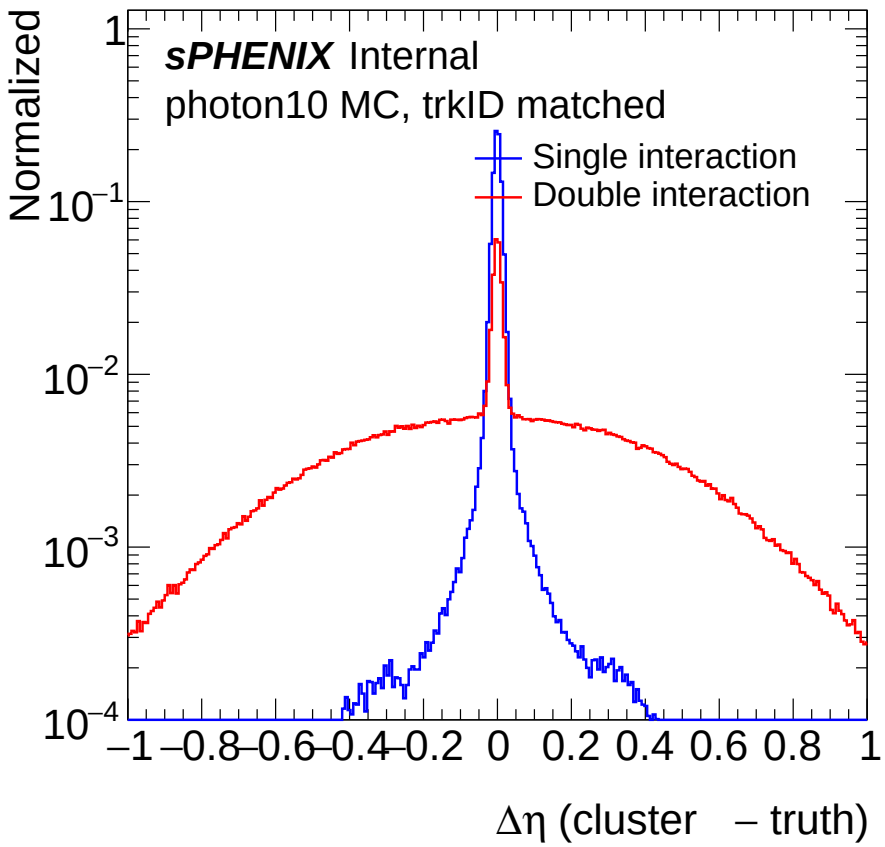


Figure 7: Normalized $\Delta\eta$ distributions for single (blue) and double (red) interaction MC. The double-interaction distribution extends well beyond $|\Delta\eta| = 0.1$ (the approximate $\Delta R$ threshold), with RMS 5× larger.

### 3.8 Impact on $\Delta R$ and $E_{\mathbf{T}}$ Matching

Table 2 summarizes the matching statistics for truth photons within the analysis acceptance ($|z_{\rm vtx}^{\rm reco}| < 60$ cm).

Table 2: Truth-matching statistics (within vertex cut, standard PPG12 truth photon selection).

| **Metric** | **Single** | **Double** |
|---|---|---|
| Truth photons selected | 476,381 | 690,450 |
| trkID matched | 465,270 (97.7%) | 635,532 (92.0%) |
| Fail $\Delta R < 0.1$ | 14,164 (3.04%) | 413,620 (65.08%) |
| Cluster $\Delta\eta$ RMS | 0.073 | 0.365 |

The $E_{\rm T}$ ratio distribution (Figure 9) shows that despite the large vertex displacement, the cluster energy measurement remains correct. The broadening of the $E_{\rm T}/p_{\rm T}^{\rm truth}$ distribution in double-interaction MC reflects the $E_{\rm T}$ recalculation at the wrong $\eta$ (due to the displaced vertex), not a degradation of calorimeter response.

### 3.9 $\Delta\eta$ Dependence on $\eta$

The $\Delta\eta$ shift depends on the cluster position in the barrel through the $\cosh\eta$ factor in Eq. (1). At $\eta = 0$, $\cosh\eta = 1$ and the shift is maximal for a given $\Delta z$. At $|\eta| = 0.7$, $\cosh\eta \approx 1.26$, reducing the shift by 20%.

Figure 10 shows the $\Delta\eta$ vs. $\eta_{\rm truth}$ correlation for both samples. In single-interaction MC, the correlation is barely visible. In double-interaction MC, the characteristic "X" pattern reflects the $\cosh\eta$ modulation.

## 4 Summary

1. The double-interaction MBD vertex is the average of two collision vertices, producing a $\Delta z$ RMS of 48.1 cm (vs. 16.9 cm for single interaction, event-level before vertex cut).

2. The reconstructed vertex distribution is actually *narrower* for double-interaction events (due to averaging), but the displacement from the truth vertex is much larger.

3. The $\Delta\eta$ shift follows the geometric prediction $\Delta\eta \approx -\Delta z/(R_{\rm CEMC} \cdot \cosh\eta)$, with cluster $\Delta\eta$ RMS broadened by a factor of 5 (0.365 vs. 0.073).

4. 74.2% of double-interaction events exceed the critical vertex displacement $|\Delta z| = 9.35$ cm, leading to a 65.1% $\Delta R$-matching failure rate for trkID-matched clusters.

5. Cluster energy response ($E_{\rm T}/p_{\rm T}^{\rm truth}$) remains centered near unity for both interaction types; the vertex shift distorts the angular position, not the energy measurement.

6. These results provide the quantitative basis for: (a) removing the geometric $\Delta R$ matching cut in favor of trkID-only truth association; (b) evaluating the pileup systematic through the double-interaction efficiency ratio; and (c) understanding the vertex-driven component of shower-shape BDT score migration.

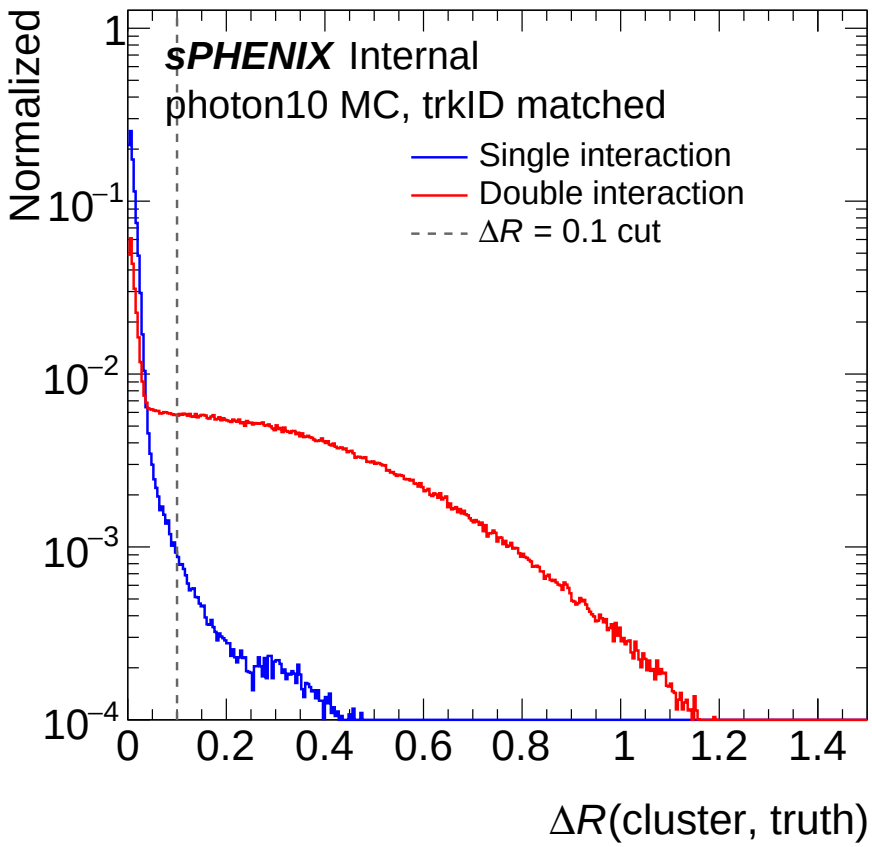


Figure 8: Normalized $\Delta R$ distributions for trkID-matched clusters in single (blue) and double (red) interaction MC. The dashed line marks the $\Delta R = 0.1$ cut. In double-interaction MC, the majority of correctly trkID-matched clusters exceed the geometric threshold.

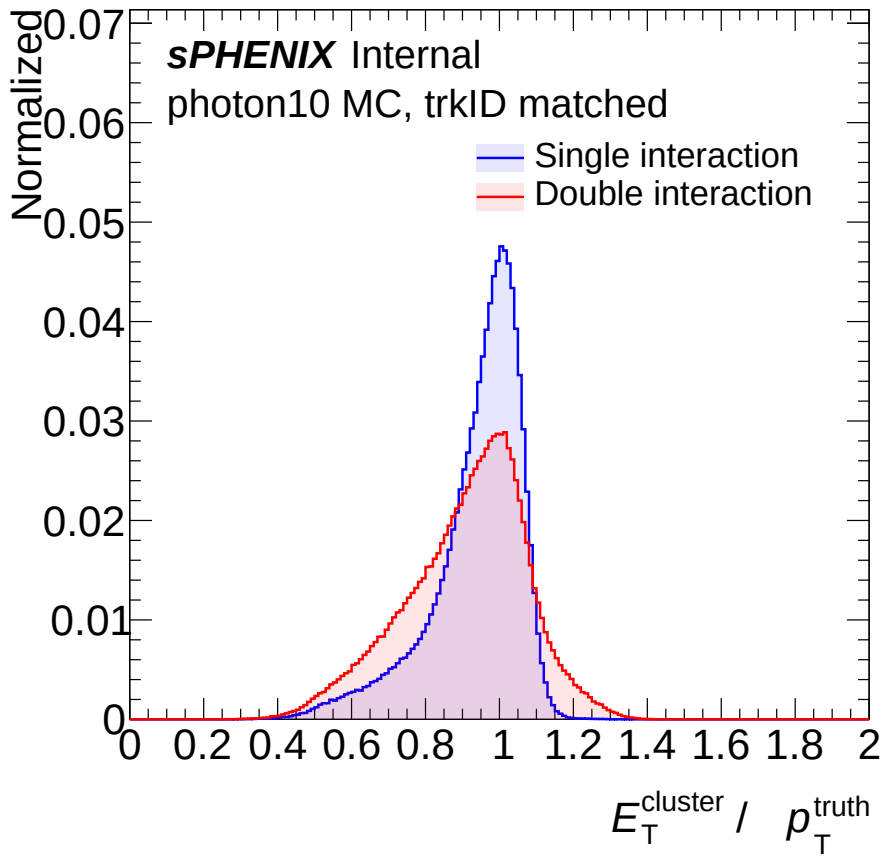


Figure 9: Distribution of $E_{\mathrm{T}}^{\mathrm{cluster}}/p_{\mathrm{T}}^{\mathrm{truth}}$ for trkID-matched clusters. The double-interaction distribution is broadened by the $E_{\mathrm{T}}$ shift from the displaced vertex, but remains centered near unity — indicating the calorimeter correctly measured the photon energy.

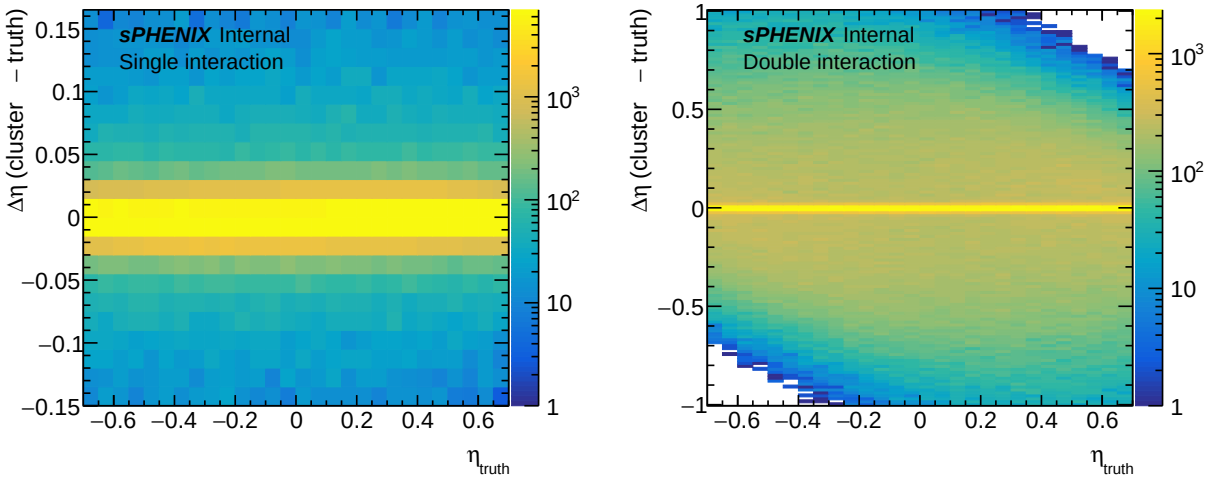


Figure 10: $\Delta\eta$ vs. truth $\eta$ for single (left) and double (right) interaction MC. Note the different $\Delta\eta$ scales. In double interaction, the spread is modulated by $\cosh\eta$, with maximum displacement at $\eta = 0$.

## Files

| File | Purpose |
|---|---|
| `efficiencytool/study_vertex_double_interaction.C` | Study macro |
| `plotting/plot_vertex_double_interaction.C` | Plotting macro |
| `results/vertex_double_interaction_study.root` | Output histograms |
| `plotting/figures/vertex_study/*.pdf` | Output figures |